

Introducing the Coxeter Flag Variety

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Summary

In prior work, BGST and Nadeau define the "Quasisymmetric Flag Variety" QFI_n which has many nice properties including:

$$H^*(\text{QFI}_n) \cong \frac{\mathbb{Z}[x_1, \dots, x_n]}{\langle f - f(0) \mid f \in \text{QSym}_n \rangle}$$

We show that this space admits a broad generalization to:

- all Lie types/Weyl groups, and
- any choice of Coxeter element geometrically realizing Coxeter-Catalan combinatorics.

Level (of generality)

	"QFI"	"type A"	full
G	gen. linear group	$GL_{n+1}(\mathbb{C})$	a reductive group (e.g. O_n)
B, B^-	triangular subgroups	$\begin{bmatrix} \mathbb{C}^* & & \\ & \mathbb{C}^* & \\ 0 & & \mathbb{C} \end{bmatrix}$	opposite Borel subgroups
T	diagonal subgroup	$\begin{bmatrix} \mathbb{C}^* & & 0 \\ & \mathbb{C}^* & \\ 0 & & \mathbb{C} \end{bmatrix}$	a maximal torus
W	the symmetric group		the Weyl group $N_G(T)/T$
$s \in S$	simple transpositions	s_i	simple reflections in W
$\beta \in \Phi^+$	(i, j) and/or $t_i - t_j, i < j$		positive roots in $\text{Char}(T)$
$\tau \in T$	all transpositions	$\tau = (i j)$	all reflections $\tau = s_\beta$ in W
c	$(1 n \dots 2)$	$(1 a_1 \dots n b_1 \dots b_k)$ $a_1 < \dots < a_{n-2-k}$ $b_1 < \dots < b_k$	$\prod_{s \in S} s$ (any order)

Flag, Schubert, Richardson varieties

The (generalized) flag variety is the coset space: G/B

In type A: $G/B \leftrightarrow \{\text{complete flags } 0 \subseteq F_1 \subseteq \dots \subseteq F_{n-1} \subseteq \mathbb{C}^n\}$

$$\begin{bmatrix} | & | & | & | \\ e_1 & e_2 & \dots & e_n \\ | & | & | & | \end{bmatrix} \leftrightarrow F_j = \text{span}\{e_i : i \leq j\}$$

Bruhat decomposition: $G/B = \bigsqcup_{w \in W} BwB/B$, $BwB/B \cong \mathbb{C}^{\text{inv}(w)}$

Schubert varieties:

- $X^w = \overline{BwB/B}$
- $X_w = \overline{B^-wB/B}$

Richardson varieties: $X_u^v = X^v \cap X_u$, nonempty only if $u \leq_{\text{Bruhat}} v$.

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c-Cambrian Congruence

Each Coxeter element determines a collection of " c -sortable" elements in W

$$\begin{aligned} 35421 &: s_2 s_1 s_3 s_4 \mid s_2 s_1 s_3 s_4 \mid s_2 s_1 s_3 s_4 \quad \checkmark \\ 53421 &: s_2 s_1 s_3 s_4 \mid s_2 s_1 s_3 s_4 \mid s_2 s_1 s_3 s_4 \quad \times \end{aligned}$$

Theorem [18]: every $w \in W$ has a unique maximum c -sortable beneath it, denoted $\pi_{\downarrow}^c(w)$.

Let $w \equiv_c v$ if and only if $\pi_{\downarrow}^c(w) = \pi_{\downarrow}^c(v)$.

QFI case: these are 312-avoiding permutations and the Sylvester congruence.

c-Noncrossing Partitions

Definition: A c -noncrossing partition in W is a

$$u = \tau_1 \tau_2 \dots \tau_k \in W \quad \text{where} \quad c = \tau_1 \tau_2 \dots \tau_k \dots \tau_n$$

Type A case: actual noncrossing set partitions

$$c = (134652) \quad \begin{array}{c} 1 \\ 2 \quad 3 \\ 5 \quad 4 \\ 6 \end{array} \quad \rightarrow \quad (146)(3)(25) \in \text{NC}(S_6, c)$$

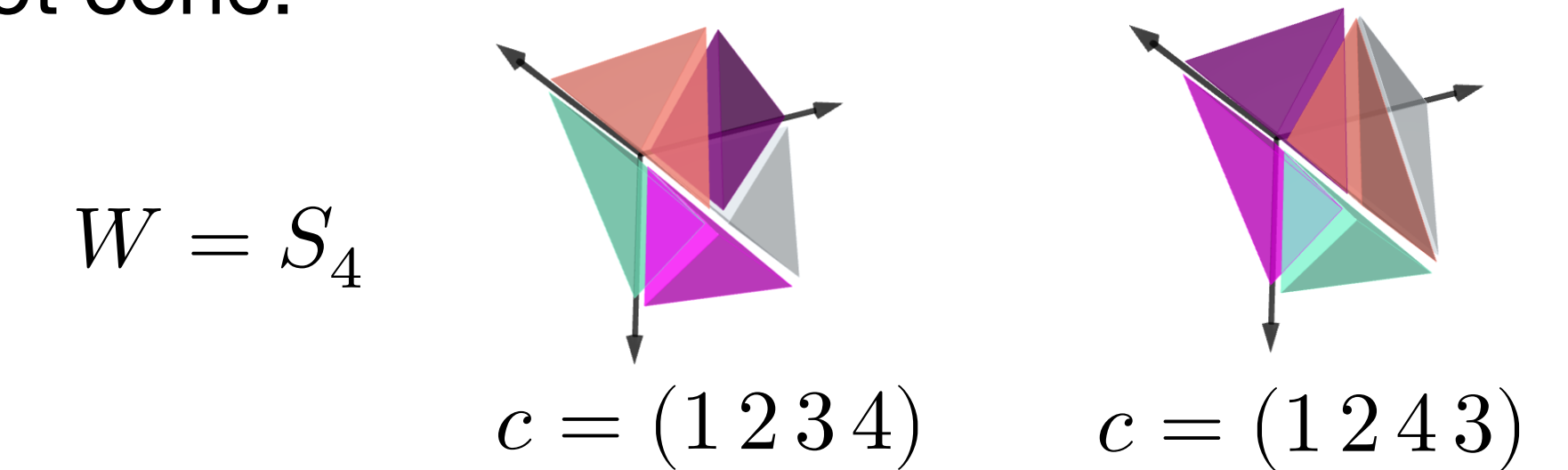
Theorem [17]: there is a bijection "cov" from c -sortable elements to $\text{NC}(W, c)$.

c-Clusters

Non-Standard Definition [5]: The positive c -Clusters are, for each $u \in \text{NC}(W, c)$:

$$C = \{\alpha \in \Phi^+ \mid s_\alpha u \in \text{NC}(W, c) \text{ and } s_\alpha u \leq_{\text{Bruhat}} u\}$$

The maximal-size positive clusters subdivide the root cone:



Coxeter Catalan:

Theorem [4, 16, 17]: Each family above is counted by the W-Catalan number $\text{Cat}(W, c) = \prod (h + d_i)/d_i$

The 'fully supported' members are counted by the *positive W-Catalan number* $\text{Cat}(W, c)^+$

Coxeter Richardson Varieties

Definition: a (c -)Coxeter Richardson variety is

$$X_w^{wc} \quad \text{for} \quad w \in W.$$

Note: $X_w^{wc} \neq \emptyset$ if and only if $\ell(wc) = \ell(w) + \ell(c)$.

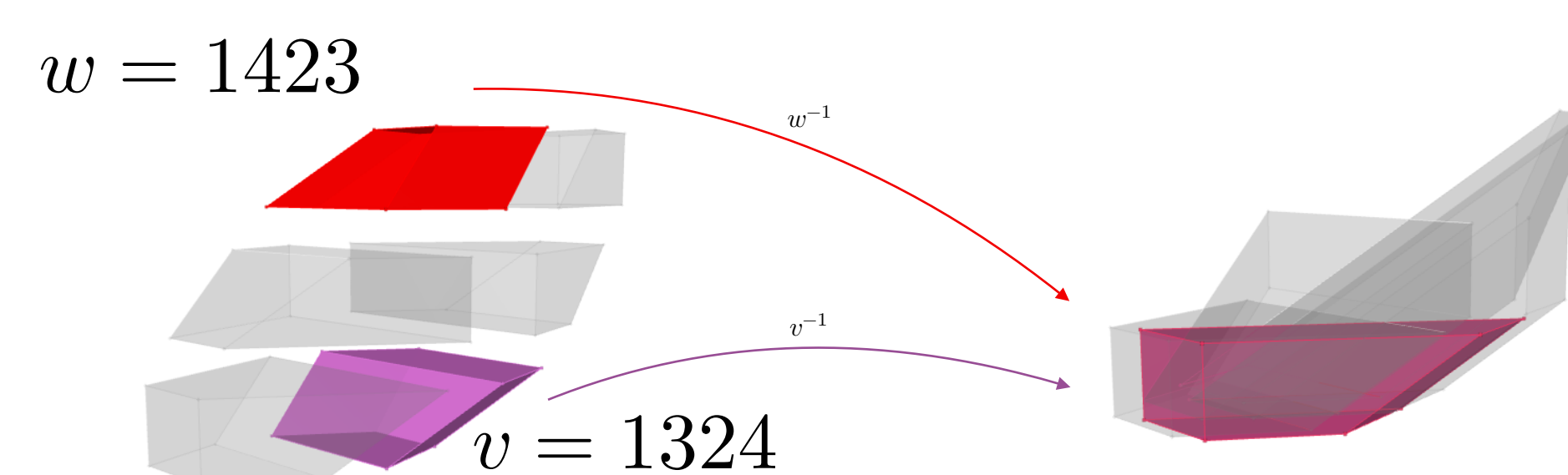
Theorem [12]: These are toric varieties whose moment polytopes subdivide the permutahedron



$$W = S_4, c = (1234) \quad W = S_4, c = (1243)$$

Strange Coincidence: for some c , Coxeter Richardsons can be equal up to translation:

$$\text{often} \quad w^{-1} X_w^{wc} = v^{-1} X_v^{vc}.$$



Classification of CR varieties

Theorem: $w^{-1} X_w^{wc} = v^{-1} X_v^{vc}$ only if

$$w_0 w^{-1} \equiv_c w_0 v^{-1}.$$

A similar result was proved in [8]; we extend it fully solve the Coxeter Richardson ambiguity.

Theorem: For $w \in W$, let $d_w = \prod_{s \leq_{\text{Bruhat}} w} s$.

Then $w^{-1} X_w^{wd_w} = v^{-1} X_v^{vd_v}$ if and only if $w \equiv_c v$.

Coxeter Flag Variety

G/B has extremal Plücker coordinates:

$$\text{Pl}_w : G/B \hookrightarrow \mathbb{P}(V^\lambda) \rightarrow \mathbb{P}(V_{w\lambda}^\lambda) \quad \text{for } w \in W.$$

Definition: The c -Coxeter flag variety is

$$\text{cfl}_c = \bigcap_{w \in W \setminus \text{NC}(W, c)} \{\text{Pl}_w = 0\} \subseteq G/B$$

c-Cluster charts

The positive cluster for $u \in \text{NC}(W, c)$ determines a *cluster chart* $\hat{X}_{\text{NC}}^u \subseteq BuB/B$

$$\hat{X}_{\text{NC}}^{4123} = \begin{bmatrix} * & 1 & 0 & 0 \\ * & 0 & * & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} B \subseteq \begin{bmatrix} * & 1 & 0 & 0 \\ * & 0 & * & 1 \\ * & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} B$$

and a *Coxeter Schubert variety*: $X_{\text{NC}}^u = \overline{\hat{X}_{\text{NC}}^u}$.

The Structure of cfl

Theorem: the Coxeter flag variety is:

1. a reducible complex of toric varieties with $\text{Cat}(W, c)^+$ -many components;
2. the union of translated Coxeter Richardsons

$$\text{cfl}_c = \bigcup w^{-1} X_w^{wc}$$

3. the *disjoint* union of cluster charts

$$\bigcup_{u \in \text{NC}(W, c)} \hat{X}_{\text{NC}}^u \quad \text{with} \quad \hat{X}_{\text{NC}}^u \cong \mathbb{C}^{s(u)}$$

Moreover, if w is c -sortable and $w^{-1} X_w^{wc} \neq \emptyset$, then

$$w^{-1} X_w^{wc} = X_{\text{NC}}^{\text{cov}(u)}$$

Graph Cohomology

Theorem: The (T -equivariant) cohomology ring for cfl_c admits a "GKM graph cohomology presentation" using the *Kreweras graph* $\text{NC} \subseteq \text{Cayley}(W)$:

$$H_T^*(\text{cfl}_c) = \left\{ (f_u)_{u \in \text{NC}} \in \mathbb{Z}[t_1, \dots, t_n]^{\oplus \text{NC}} \mid f_u - f_{s_\alpha u} \in (\alpha) \text{ for all edges} \right\}$$

which is a free module of rank $\text{Cat}(W, c)$.

Corollary: For fixed G and distinct $c, c' \in W$, $H^*(\text{cfl}_c) \cong H^*(\text{cfl}_{c'})$, even though $\text{cfl}_c \not\cong \text{cfl}_{c'}$.

Corollary: For $G = \text{GL}_{n+1}$ and $wcw^{-1} = (1 n \dots 1)$,

$$H^*(\text{cfl}_c) \cong \frac{\mathbb{Z}[x_1, \dots, x_{n+1}]}{\langle f(x_{w(1)}, \dots, x_{w(n+1)}) - f(0) \mid f \in \text{QSym}_{n+1} \rangle}$$

